



Parktown Boys' High School

Geography Department

Grade 12 Mapwork Test

Examiner: Mr. D Grace
Moderator: Mr. A Meintjes

Time: 1 Hour
Total: 60 Marks

Instructions:

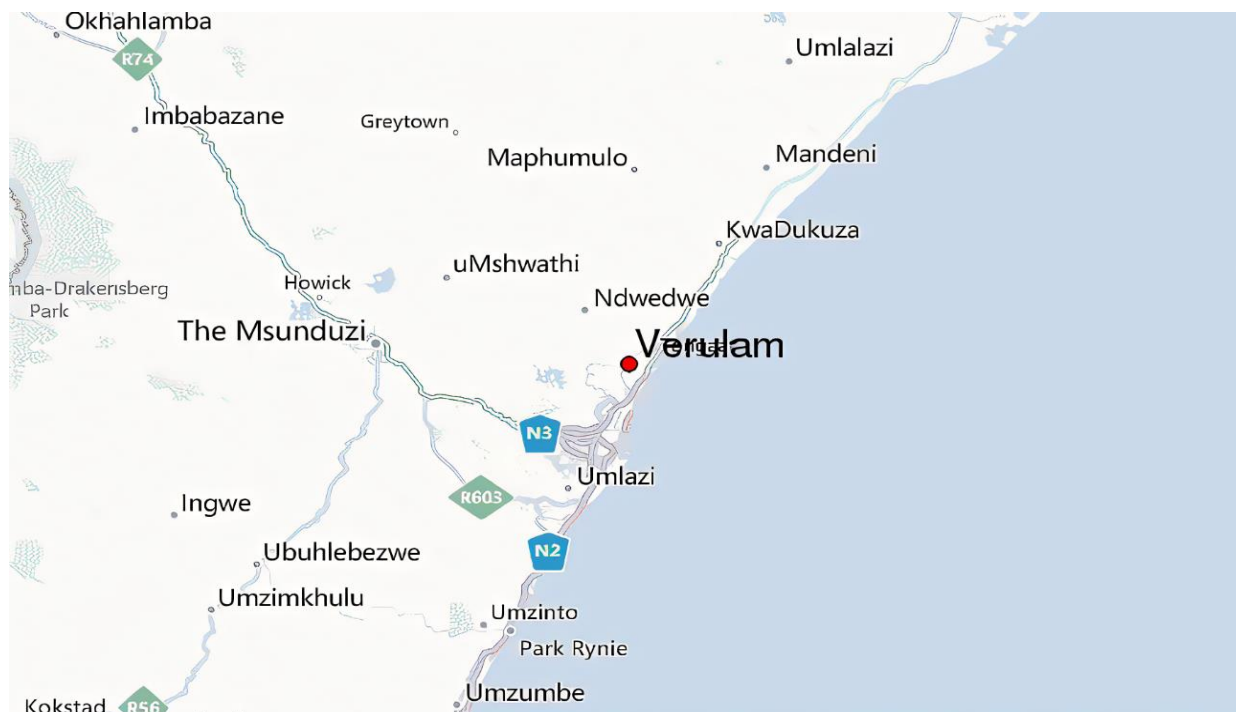
1. Make sure you put your **name** and **class teacher** on the top of your answer sheet
2. Answer **ALL** the questions on your answer sheet
3. Number the answers correctly according to the numbering system used in this question paper.
4. All diagrams must be at least 7 – 10 lines in size and fully labelled
5. Write neatly and legibly
6. GOOD LUCK!

RESOURCE MATERIAL

1. An extract from the topographic map 2931 CA VERULAM
2. Orthophoto map 2931 CA 11 VERULAM

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions in the spaces provided in this question paper.
2. Show ALL calculations and formulae, where applicable. Marks will be allocated for these.
3. Indicate the unit of measurement or compass direction in the final answer of calculations, e.g. 10 km; 2,1 cm; west of true north.
4. You may use a non-programmable calculator.
5. The area demarcated in RED on the topographic map represents the area covered by the orthophoto map.
6. The following English terms and their Afrikaans translations are shown on the topographic map:

BACKGROUND INFORMATION ON VERULAM

Verulam is a town 28 kilometres north of Durban in KwaZulu-Natal, South Africa and is now part of the eThekweni Metropolitan Area. In 1850 a party of 400 Methodists settled here and formed the town. The town was then named after the Earl of Verulam, patron of the Brits who settled here. It is now a town inhabited mainly by Indians.

[Source: [https://en.wikipedia.org/wiki/Verulam,_KwaZulu](https://en.wikipedia.org/wiki/Verulam,_KwaZulu_Natal) Natal]

QUESTION 1 – MULTIPLE CHOICE AND CALCULATIONS

1.1 The questions below are based on the 1: 50 000 topographic map 2931 CA VERULAM as well as the orthophoto map 2931 11 VERULAM. Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) in the block next to each question. (4 x 1) (4)

1.1.1 The province located west of VERULAM is ...

- A. Gauteng.
- B. Limpopo.
- C. North West.
- D. Free State

1.1.2 The contour interval on the topographical map is ... metres.

- A. 5
- B. 10
- C. 20
- D. 25

1.1.3 The map index of the topographic map, south of 2931 CA VERULAM, is ...

- A. 2931 CC
- B. 2931 CD
- C. 2931 AC
- D. 2931 AA

1.1.4 The coordinates of Mount Moreland in block **D4** on the topographic map is ...

- A. 29°38'12" E 31°05'30" S
- B. 29°40'00" S 29°02'35" E
- C. 29°38'12" S 31°05'30" E
- D. 29°39'45" E 31°06'56" S

1.2 Refer to point G in block **B5** and point F in block **C5** on the topographic map.

- 1.2.1 Using the information on the topographic map, determine the magnetic bearing from point G in block **B5** to point F in block **C5** on the topographic map for the current year. Show ALL calculations. Mark will be awarded for calculations. (7 x 1) (7)

Formula: MB = True Bearing (TB) + Magnetic Declination (MD)

- 1.2.2 Why is it necessary to calculate the current magnetic declination? (1 x 2) (2)

- 1.2.3 Refer to the trig beacon 70 in block **C3** and spot height 118 in block **B2** on the topographic map.

- (a) Calculate the average gradient between trig beacon 70 in block **C3** and spot height 118, in block **B2** on the topographic map.. (5 x 1) (5)

Formula: Average gradient = $\frac{\text{Vertical interval (VI)}}{\text{Horizontal equivalent (HE)}}$

- (b) Is there intervisibility between trig beacon 70 in block **C3** and spot height 118 in block **B2**? (2 x 1) (2)

Yes or No: _____

Give a reason for your answer:

QUESTION 2 - APPLICATION AND INTERPRETATION

2.1 The questions below are based on the 1: 50 000 topographic map 2931 CA VERULAM as well as the orthophoto map 2931 CA 11 VERULAM. Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) in the block next to each question.

2.1.1 The natural feature point J in block **D4** is a..... (1 x 1) (1)

- A. Windpump.
- B. Cultivated land.
- C. Non-perennial water.
- D. Reservoir.

2.1.2 The non-perennial river flowing in block **D4** on the topographic map is flowing in a ... direction.

- A. Western
- B. Southerly
- C. Northerly
- D. Easterly

(1 x 1) (1)

2.1.3 The Area calculation for the RED demarcated area of the orthophoto on the topographic map is ... in km² (1 x 1) (1)

- A. 1.5 KM
- B. 2.7 KM
- C. 2.1 KM
- D. 1.7 KM

2.2 Refer to settlements in block **D1** and in block **C5** on the topographic map.

2.2.1 Identify the settlement types in **blocks C4** and **D1** respectively. (2 x 1) (2)

C4: _____

D1: _____

2.2.2 The average temperatures of the settlement in block **D1** are higher than the settlement in block **C4**. What is this climatic phenomenon called? (1 x 1) (1)

2.2.3 Why does settlement in block **D1** records higher temperatures? (1 x 2) (2)

- 2.2.4 What strategies can the municipality of Durban (eThekweni) implement to reduce the impact of this climate phenomenon (mentioned in question 2.2.2) (1 x 2) (2)

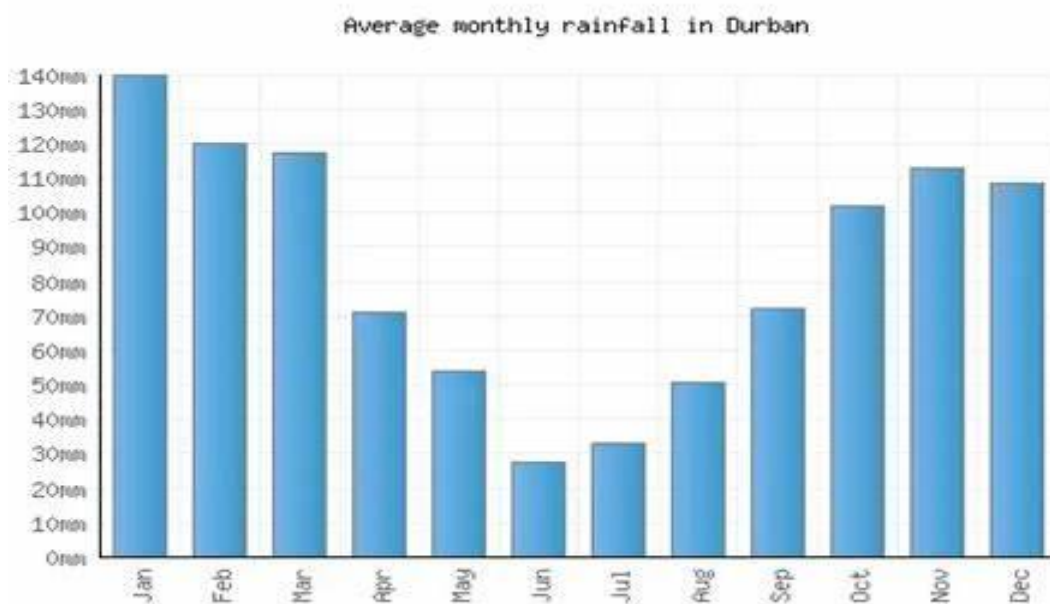
2.3 The western parts of Verulam experience valley winds at night.

- 2.3.1 Name the type of valley winds that affect the western parts of Verulam at night. (1 x 1) (1)

- 2.3.2 Katabatic winds are associated with temperature inversion. What is temperature inversion? (2 x 1) (2)

- 2.3.3 Define temperature inversion? (1 x 2) (2)

2.4 Refer to the graph below, showing the average monthly rainfall in Verulam.



- 2.4.1 The mapped area (on the topographic map) receives seasonal rainfall. Give two points of evidence from the topographic map to support this statement. (2 x 1) (2)

- 2.4.2 Provide the month during which Verulam received the lowest rainfall. (1 x 1) (1)

- 2.4.3 The northern part of Verulam is prone to flooding. Suggest measures evident on the topographic map (in block **B1**) to reduce the risk of flooding. (1 x 2) (2)

- 2.5 Refer to the passage and the picture below:

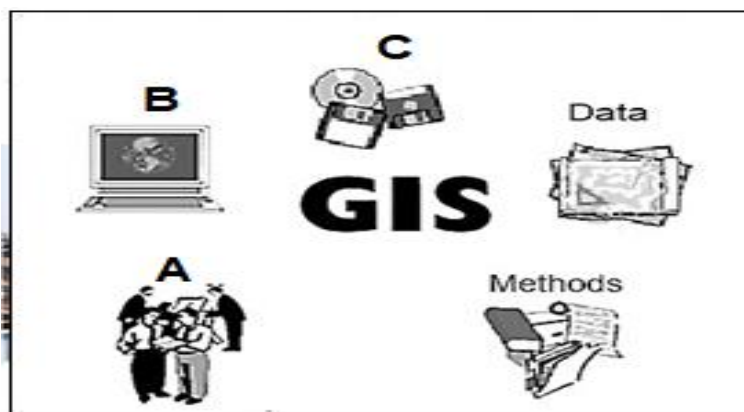


- 2.5.1 Due to pollutants shown above, Verulam experiences the effect of pollution dome. Define the concept pollution dome. (1 x 1) (1)

- 2.5.2 Mention TWO solutions that the local municipalities can implement to reduce the effect of pollution dome in Verulam. (2 x 2) (4)

QUESTION 3 – GEOGRAPHIC INFORMATION SYSTEMS

3.1 The following diagram shows the components of GIS.



3.1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A – D) in the box next to each question. (2 x 1) (2)

(a) Component **A** in the illustration above is...

- A software
- B hardware
- C people
- D processes

(b) Which of the following can be seen as a layer...

- A. Elevation
- B Temperature
- C Water
- D All of the above

3.2 Refer to block **B4** on the topographic map.

3.2.1 Differentiate between the raster and vector data. (2 x 1) (2)

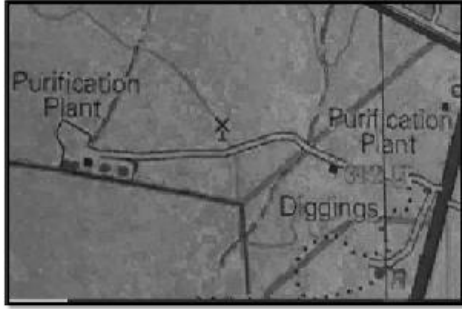
3.2.2 Identify a line feature in block **E1**.

(1 x 1) (1)

Line :

3.2.3 Refer to the image, of a portion of the topographic map, below and draw the symbol of the point feature that is used to extract ground water

(1 x 1) (1)

Image	Drawing
	

3.2.4 What is the importance of the point feature identified in Question 3.2.3 to the cultivated land in this area?

(1 x 2) (2)

3.3 Refer to diagram **A** and diagram **B** below.

Diagram A:	Diagram B:
	

3.3.1 Resolution refers to the detail to which a map depicts the location and shape of geographic features. Which image **A** or **B**, has a better resolution?

(1 x 1) (1)

3.3.2 Give a reason for this difference in resolution between Image **A** and Image **B**.

(1 x 2) (2)

3.4 High traffic volumes and congestion on the main road 102 in block **D1** and **E1** on the topographical map map have been reported.

3.4.1 Identify the type road that joins the main road 102 in block **B4/D1** on the topographical map. (1 x 1) (1)

3.4.2 Suggest THREE ways in which the eThekweni municipality can use GIS to reduce traffic congestion in this area.

(3 x 1) (3)

(a) _____

(b) _____

(c) _____

TOTAL : 60

ROUGH WORK